

BIOLOGY — GRADE 11 | GROWTH AND DEVELOPMENT IN PLANTS

Quick Check

Lesson 2: Growth vs Development and Inside the Seed: Maize H614 and Rosecoco Beans

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 2

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On Mr. Wafula's farm, a bean plant makes its first single-bladed (unifoliate) leaves on Day 12 and three-part (trifoliate) leaves on Day 18. How should this change be classified?

- A** Growth only, because the plant has become bigger
- B** Development, because the leaf changes in form and becomes more complex through differentiation
- C** Growth only, because there are now more leaves
- D** Neither growth nor development, because leaves are not part of the seed

A dry maize grain swells after 25 mm of rain falls on the field. Why is this swelling NOT counted as true growth?

- A** Swelling makes new organs form, so it is development, not growth
- B** Growth only means getting taller, and the grain did not get taller
- C** Maize grains are not living things, so they cannot grow
- D** Water can be lost again, so the increase is not a permanent, irreversible increase in size or dry mass

The stem of a maize plant is 6 mm wide on Day 14 and 22 mm wide on Day 42. By how much did the stem width increase, and is this growth?

- A** 16 mm; it is growth, because the size increased permanently
- B** 28 mm; it is growth, because the stem became thicker
- C** 16 mm; it is not growth, because the plant did not get taller
- D** 22 mm; it is development, because the stem changed form

A learner splits a soaked Rosecoco bean seed into two halves. Which part of the bean stores the food that the young plant will use?

A The testa

B The two cotyledons

C The micropyle

D The hilum

In a maize grain, which part protects the shoot tip (plumule) as it pushes up through the soil?

A The coleoptile

B The coleorhiza

C The scutellum

D The endosperm

Which statement correctly describes how a maize grain (a monocot) gets food for its embryo?

- A** Food is stored in two large cotyledons, as in a bean
- B** The endosperm is the embryo, and it is the part that grows into the plant
- C** Food is stored in the endosperm, and the single cotyledon (scutellum) absorbs it for the embryo
- D** Maize has no cotyledon, so the embryo makes all its own food

A learner revises their Day 1 model by adding labelled seed drawings and arrows inside the soil. Which arrows show the correct development of the embryo parts?

- A** Radicle becomes the leaf; plumule becomes the root
- B** Radicle becomes the root; plumule becomes the shoot
- C** Radicle becomes the shoot; plumule becomes the root
- D** Radicle becomes the testa; plumule becomes the cotyledon

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).